

TECHNICAL PORTFOLIO

Engineering Case Studies & System Architecture

A curated survey of production systems designed and shipped across food & hospitality, real estate, AI automation, platforms, and agency delivery — with the architecture and algorithmic logic behind each build.

NEXT.JS

SUPABASE

NODE / EXPRESS

THREE.JS

LLM ORCHESTRATION

RAG & VECTOR SEARCH

OVERVIEW

Eleven Builds. Five Domains. One Engineering Standard.

Designing and shipping production-grade software across four key verticals—hospitality, real estate, AI automation, and applied platforms — and operates its own agency engine to deliver the same rigor to client work. Every project below is documented at the architecture level: the stack decisions, the core algorithmic flow, and the reasoning behind each trade-off.

11

PRODUCTION SYSTEMS

5

DOMAINS COVERED

9+

CORE TECHNOLOGIES

1

DELIVERY STANDARD

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01

Food & Hospitality Ordering Systems

Real-time ordering, booking, and multi-tenant delivery infrastructure built for regional hospitality brands — engineered for speed, reliability, and effortless replication across new vendors.

Konaseema Vantillu

Mannat Lodge

Veggies Express

Konaseema Vantillu

“Real-time flavor, zero-latency fulfillment.”

Next.js

Supabase

PostgreSQL Realtime

Server Actions

USE

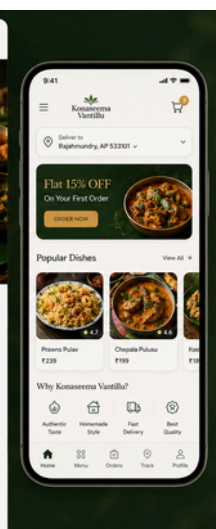
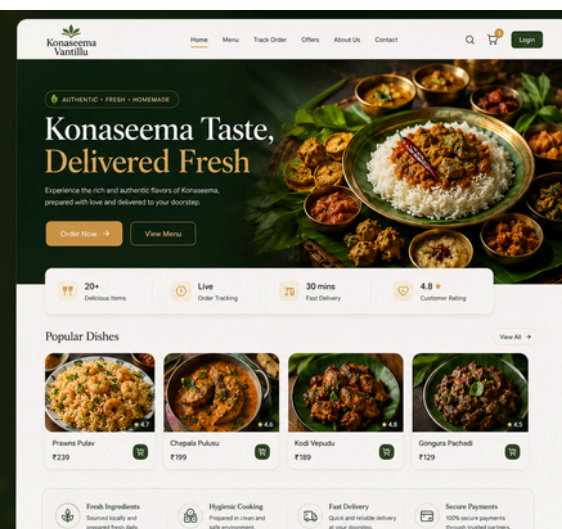
A live, high-performance food ordering platform tailored for regional cuisine, featuring instant order tracking for customers and real-time management tooling for kitchen staff.

ARCHITECTURE

Built on a Next.js frontend for rapid page rendering and strong SEO. The backend runs on Supabase, leveraging real-time PostgreSQL listeners in place of a custom WebSocket layer — cutting infrastructure overhead while keeping every screen instantly in sync.

ALGORITHM & CORE FLOW

- Order Placement** — customer submits cart data via a Next.js Server Action.
- Database Mutation** — the order is written to the Supabase orders table with status pending.
- Real-Time Broadcast** — the kitchen dashboard, subscribed via Supabase Realtime (`supabase.channel().on('postgres_changes')`), renders the new order card instantly, with no page refresh.
- State Transition** — kitchen staff move the order from cooking to dispatched; the customer's UI updates automatically through the same real-time listener hook.



UI/UX
DESIGN

BRAND
KIT

Mannat Lodge

“Robust backends for seamless hospitality.”

Node.js

Express.js

Relational DB

Transaction Locking

USE

A comprehensive property and hospitality management system built to handle heavy server-side logic, room bookings, and guest administration.

ARCHITECTURE

Powered by a custom Express.js (Node.js) backend for maximum routing control and middleware flexibility, paired with a relational database engineered to manage complex, multi-table transactional logic.

ALGORITHM & CORE FLOW

1

Booking Validation — on checkout, a custom Express middleware checks room availability using a date-range collision algorithm:

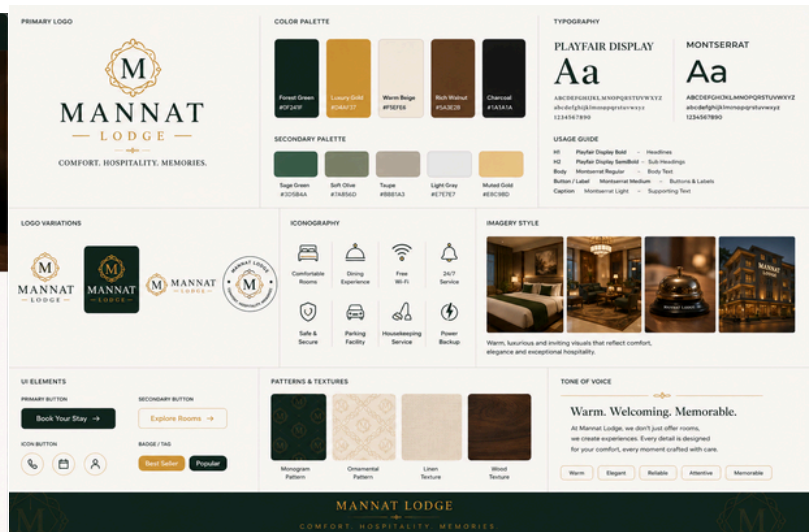
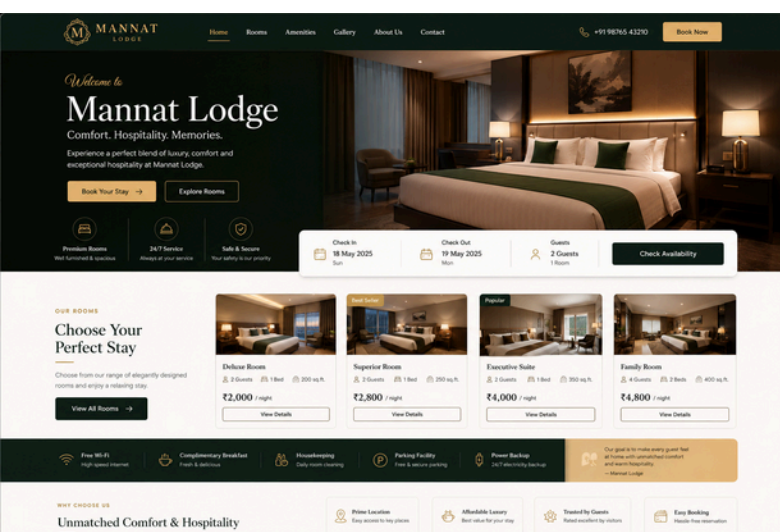
$$\text{Collision} = (\text{CheckIn} < \text{ExistingCheckOut}) \wedge (\text{CheckOut} > \text{ExistingCheckIn})$$

2

Transaction Lock — if the room is free, a database transaction locks the room row, preventing double-bookings.

3

Confirmation — the server generates a unique booking token and updates guest metrics.



UI/UX DESIGN

BRAND KIT

Veggies Express

"The blueprint for scalable micro-delivery."

Vite

React

Multi-Tenant SaaS

Dynamic Config

USE

A lightning-fast, lightweight delivery site blueprint, structured explicitly to serve as a reusable, white-label multi-tenant SaaS foundation.

ARCHITECTURE

Developed with Vite + React for an ultra-lean client bundle. The architecture emphasizes modular, environment-driven configuration so the same codebase can be spun up for new vendors with minimal code changes.

ALGORITHM & CORE FLOW

1

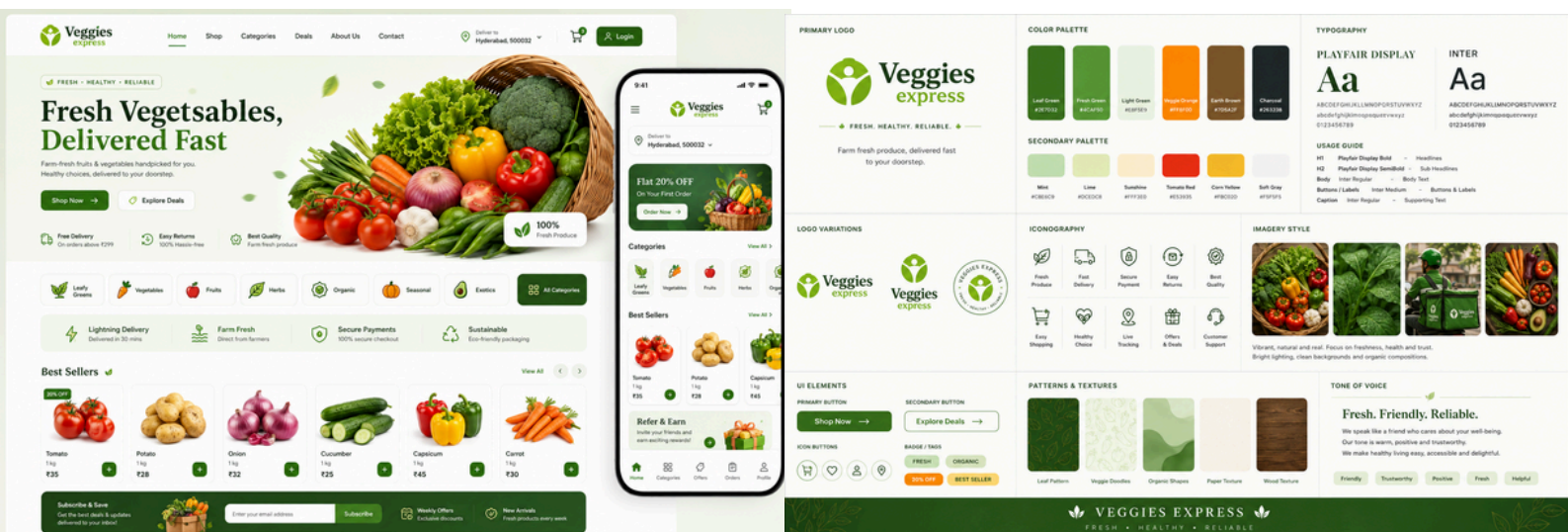
SaaS Injection — the application reads a vendor ID or subdomain from the URL host string.

2

Context Hydration — a global context hook fetches config maps (branding, catalog, pricing rules) scoped strictly to that vendor tenant.

3

Dynamic Routing — Vite renders the catalog grid against the resolved vendor context, so every deployment feels purpose-built while sharing one shared codebase underneath.



UI/UX
DESIGN

BRAND
KIT



02

RealEstate & 3D Visualization

Property discovery platforms, rapidly rebrandable listing templates, and browser-native 3D walkthroughs that bring floor plans to life for buyers before they ever set foot onsite.

Datta Properties Vizag

Royal Baltic Clone

Flatc 3D Viewer

Datta Properties Vizag

"Connecting local spaces with modern experiences."

Next.js

SSG

ISR

Indexed Query Layer

USE

A production-ready, client-facing real estate discovery platform focused on property listings across the Visakhapatnam region.

ARCHITECTURE

Built on Next.js, combining Static Site Generation for marketing pages with Incremental Static Regeneration to keep property inventory fresh — without sacrificing page load speed.

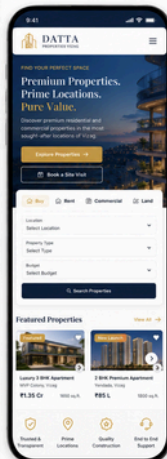
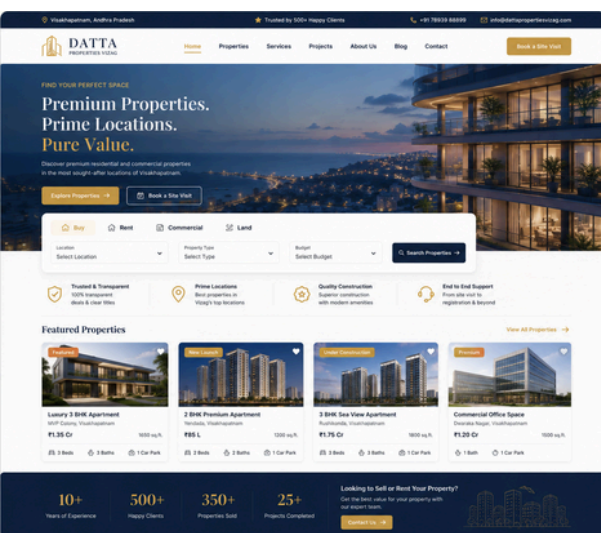
ALGORITHM & CORE FLOW

1

Query Processing — users enter search parameters such as location, budget, and BHK.

2

Cache Interception — Next.js serves the pre-rendered page when available; on a cache miss, a structured query hits indexed database columns to return results in milliseconds.



UI/UX
DESIGN

BRAND
KIT

Royal Baltic Clone

Next.js

Component-Driven UI

Config-Based Theming

USE

A high-end real estate UI template designed as an easily adaptable asset for pitching luxury real estate developers.

ARCHITECTURE

Built inside a component-driven Next.js framework using reusable UI primitives, layout skeletons, and structured dummy schemas.

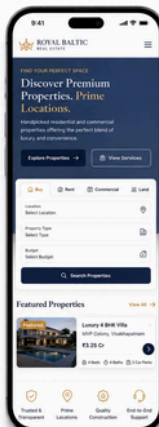
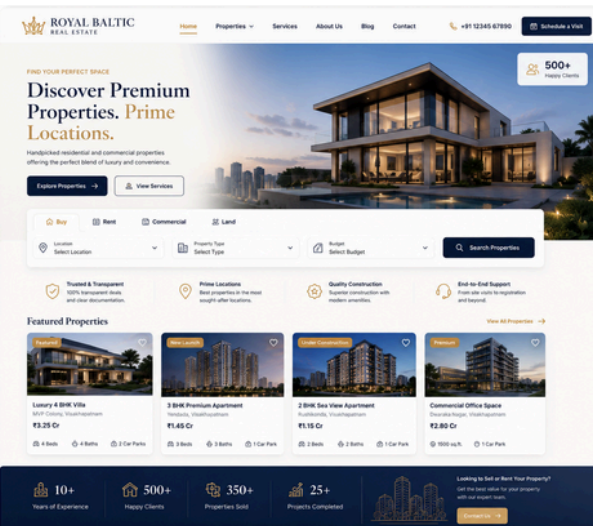
ALGORITHM & CORE FLOW

1

Schema Mapping — every UI block pulls its structural metadata from a central configuration file.

2

Rapid Rebranding — editing a single configuration file updates color variables, image assets, and listing details across the entire application instantly.



UI/UX
DESIGN

BRAND
KIT

Flatc 3D Viewer

Three.js

WebGL

GLTF / GLB

Raycasting

USE

An interactive PropTech visualization engine enabling potential buyers to explore 3D architectural floor plans directly in their web browser.

ARCHITECTURE

Built on Three.js / WebGL, embedded within a frontend container to deliver hardware-accelerated 3D graphics without requiring external plugins.

ALGORITHM & CORE FLOW

1

Asset Loading — the browser asynchronously requests low-poly 3D models in GLTF / GLB format.

2

Scene Initialization — the renderer builds camera perspective, ambient lighting, and raycasting layers.

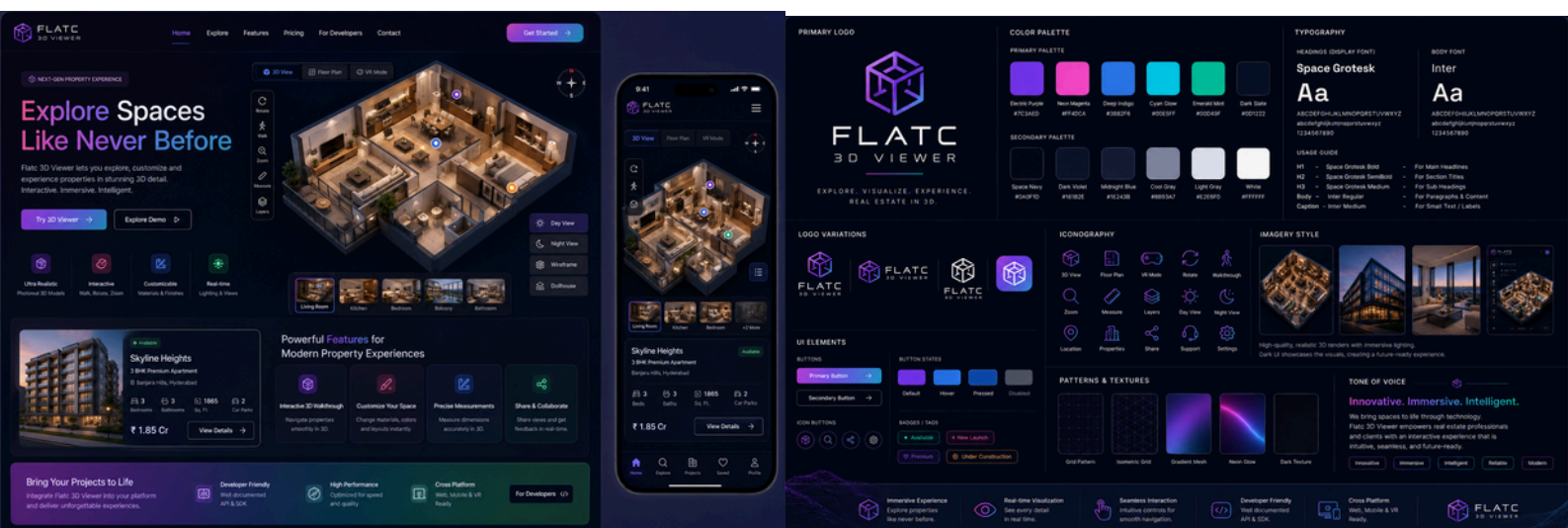
3

Raycasting Selection — a click maps a mouse-vector to the exact 3D coordinate intersection:

$$\hat{v}_{ndc} \rightarrow \text{Raycaster} \rightarrow \text{Mesh Intersection}$$

4

State Interaction — the selected room is highlighted, with real-time square footage and pricing surfaced alongside it.



**UI/UX
DESIGN**

**BRAND
KIT**



03

AI & Automation Ecosystems

Agentic systems and conversational AI that operate across telephony,

messaging, and campus knowledge bases — orchestrating LLMs against real operational and institutional data.

Maya

LifeOS

College AI Chatbot

Maya

"High-volume care, localized voice, global scale."

Event-Driven Core

Baileys (WhatsApp)

SIP Telephony

LLM Orchestration

Express / Next.js

USE

A multilingual, high-concurrency automated customer service agent running real-time support over telephony and messaging channels for travel operations.

ARCHITECTURE

Built on a decoupled, event-driven core to absorb high traffic — linking telecommunication lines, an automated WhatsApp layer via Baileys, large language models, and an Express/Next.js monitoring dashboard.

ALGORITHM & CORE FLOW

1

Inbound Webhook — a text or voice message reaches the backend gateway via WhatsApp or a SIP provider.

2

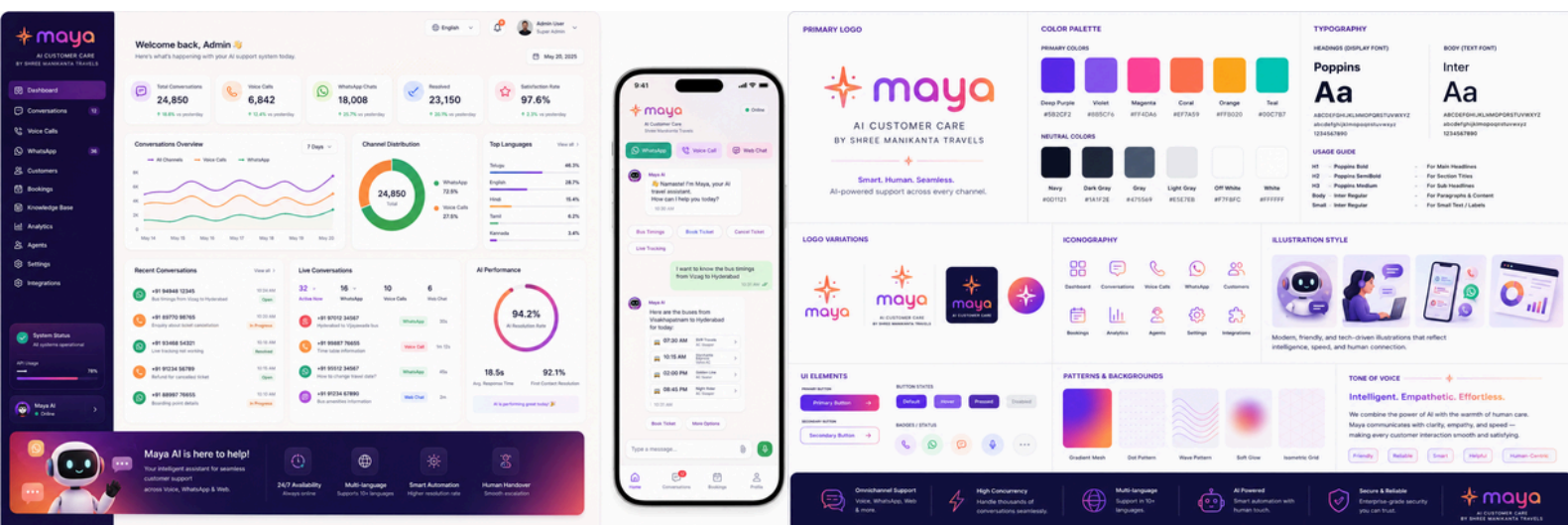
Context Retrieval — the system pulls historical customer logs and matching travel route tables.

3

Intent Extraction — the message is passed through an LLM orchestration layer to determine intent (cancel ticket, check availability, and so on).

4

Action Execution — once validated, the system runs an automated booking update and converts the reply into natural-sounding speech or formatted text for the user.



UI/UX DESIGN

BRAND KIT

LifeOS

Agentic Framework

Multi-Agent Orchestration

Vector Store

USE

A conceptual productivity “super app” that deploys autonomous AI agents to manage calendars, track routines, and execute tasks across different user age groups.

ARCHITECTURE

Powered by an agentic framework (LangChain-style or a custom multi-agent loop) operating on top of a secure, personal vector-store database.

ALGORITHM & CORE FLOW

1

Ingestion Loop — the user states a broad goal, e.g. “organize my day for exam prep.”

2

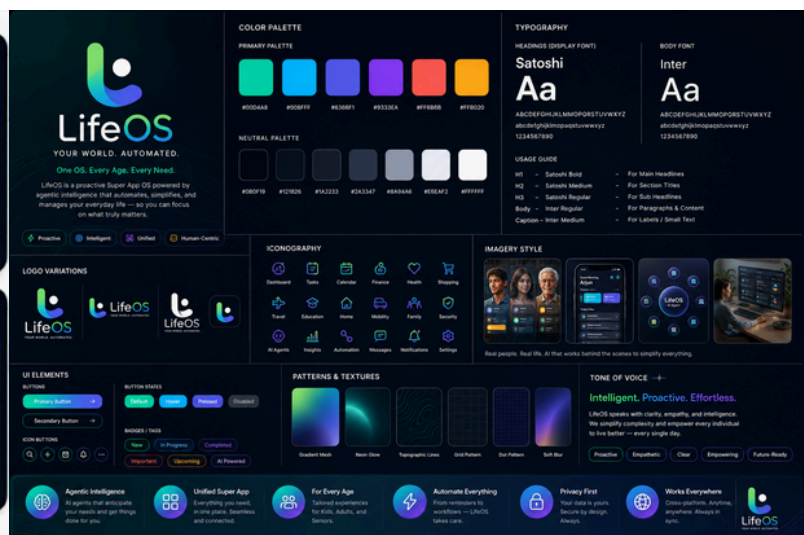
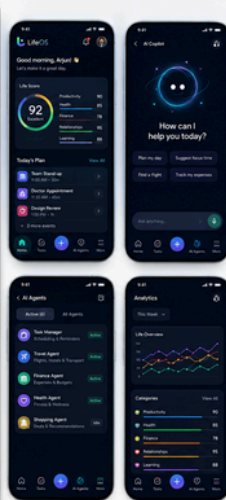
Task Decomposition — a reasoning engine breaks the goal into clear sub-tasks.

3

Execution Routing — specialized sub-agents take specific tasks: cross-checking calendars, setting focus blocks, drafting study reminders.

4

Adaptive Feedback — the schedule updates dynamically as tasks are completed or delayed through the day.



UI/UX DESIGN

BRAND KIT

College AI Chatbot

"Campus intelligence, instantly accessible."

RAG

Vector Embeddings

pgvector / Pinecone

LLM

USE

A specialized conversational AI chatbot that helps students quickly navigate college resources, admissions information, and academic regulations.

ARCHITECTURE

A production-grade Retrieval-Augmented Generation (RAG) deployment, engineered so answers stay factual and are pulled directly from official college documentation.

ALGORITHM & CORE FLOW

1

Embedding Stage — college PDFs, FAQs, and rulebooks are chunked and converted into vector embeddings.

2

Vector Search — a student's question is vectorized and matched against a database (Pinecone or pgvector)

Similarity =

$$\frac{A \cdot B}{|A| |B|}$$

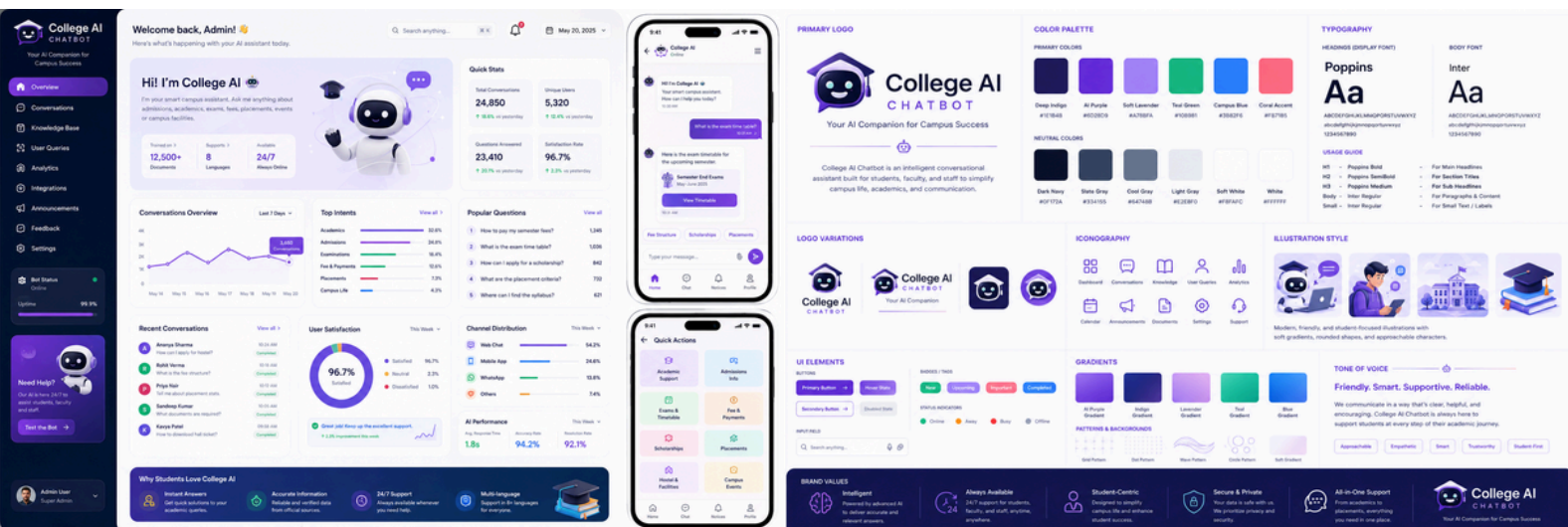
using cosine similarity:

3

Prompt Augmentation — the most relevant text chunks are inserted directly into the LLM's context window.

4

Generation — the LLM produces an accurate, hallucination-free answer grounded only in the provided context.



UI/UX
DESIGN

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KIT



04

Platforms & Applications

Purpose-built applications that connect people to opportunity — combining data-driven matching engines with rich, visual interfaces.

Optic

Optic

Matching Engine

3D UI

Admin Dashboard

USE

An interactive networking and placement platform built to connect students directly with targeted corporate internship opportunities.

ARCHITECTURE

A rich application platform featuring 3D elements, profile match-scoring metrics, and centralized administrative controls for university placement cells.

ALGORITHM & CORE FLOW

1

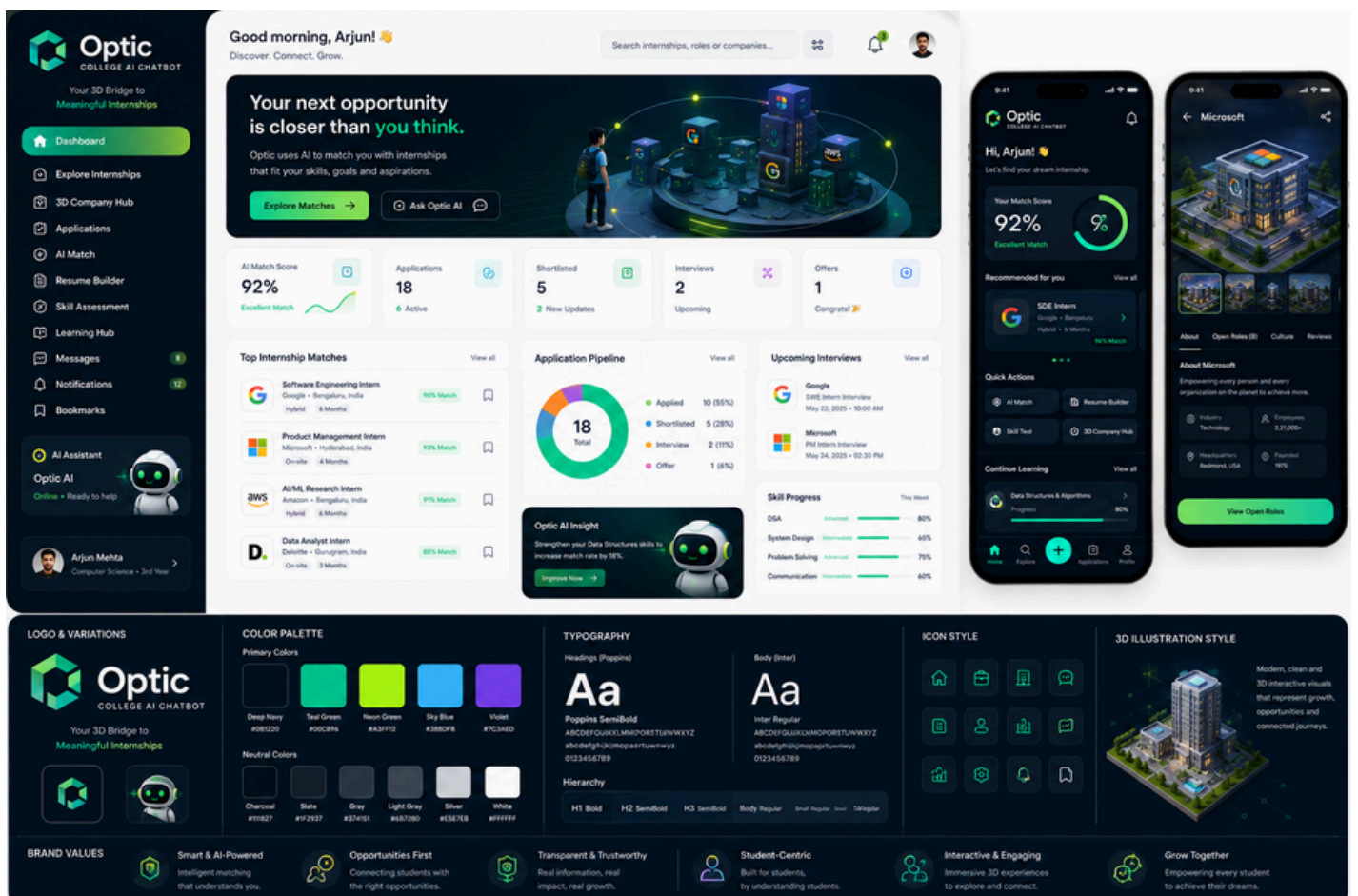
Profile Quantization — student skillsets, project tags, and academic performance are converted into structured data vectors.

2

Matching Engine — an algorithm scores the compatibility between student profiles and employer requirements.

3

Dynamic Delivery — high-scoring matches surface on the student's dashboard through interactive visual states that encourage quick application.





05

Agencies & Digital Ventures

The delivery engine behind every build in this portfolio — turning proven, reusable components into fast, secure web solutions for regional businesses.

GDS Media Lab

GDS Media Lab

Next.js

Vite

Node.js

Rapid Prototyping

USE

The central umbrella agency engine providing full-stack web engineering, custom UI/UX transformations, and results-driven digital marketing strategy for regional businesses.

ARCHITECTURE

An agile agency framework built on modern toolchains — Next.js, Vite, Node.js — paired with rapid prototyping tools to deliver fast, secure web solutions for every client engagement.

ALGORITHM & DELIVERY PIPELINE

1

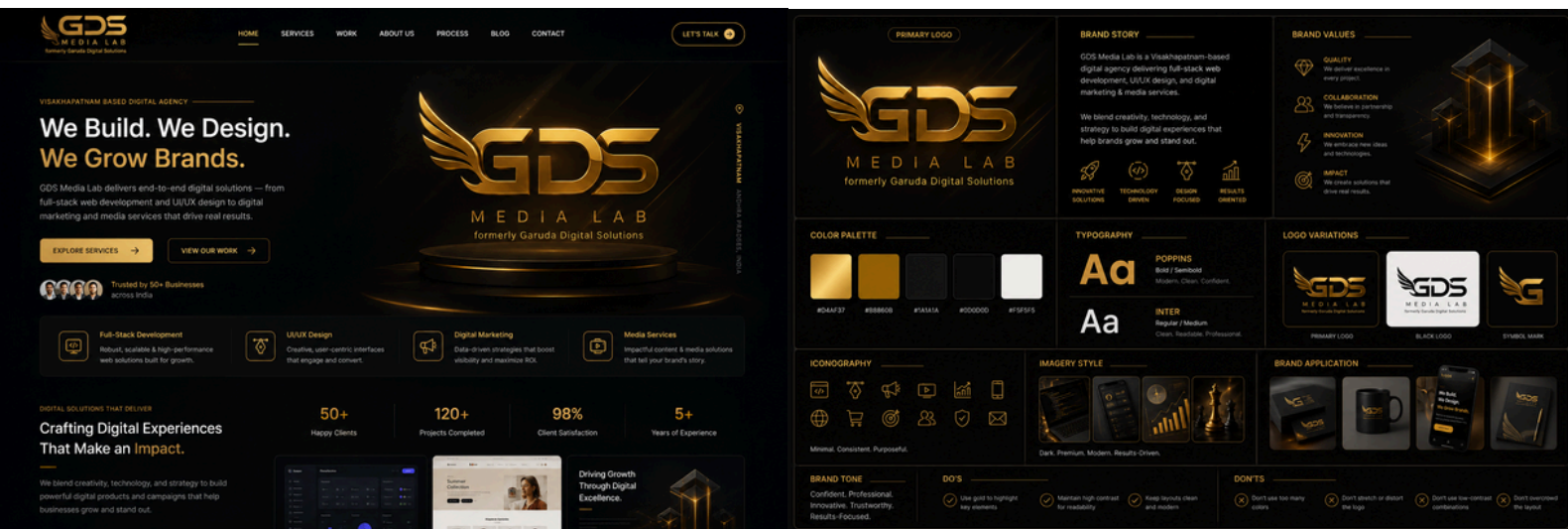
Discovery & Auditing — the agency maps a business's needs to choose the right stack: Next.js for SEO-heavy sites, Vite for internal utility tools.

2

Modular Production — engineering teams reuse pre-built boilerplate components, drawn from projects like Veggies Express and Royal Baltic Clone, to sharply cut deployment timelines.

3

Optimization Loop — every project clears an automated checklist covering core web vitals, accessible layouts, and reliable database indexing before it ships.



UI/UX
DESIGN

BRAND
KIT

Built on real architecture. Proven in **production.**

Each system in this portfolio was designed to solve a specific operational problem — then engineered to scale beyond it. This elevated benchmark of digital innovation serves as the foundation for every collaboration.